

LEDGER · 2026-09-01 · MEMBRANE SCIENCE & WATER TREATMENT

Supramolecular Membrane Rejuvenation Kits via Metal-Phenolic Network Deposition

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Commercial Polyurethane Adhesive Sealants (e.g. 3M 550 Polyurethane, Henkel Loctite Hysol)
Entry application	In-situ restoration and blister-prevention of degraded spirally-wound NF and RO membrane elements in mid-scale industrial water treatment facilities
Measured advantage	>13x improvement in treated area water permeability (13.73-21.66 LMH/bar vs ~0 LMH/bar for polyurethane sealant)
Time to first revenue	6 months
Gross margin at parity	88.8%
Startup CapEx	\$145,000
Payback from first sale	0.3 months

Capital productivity

59.59x

Regulatory risk

Moderate — qualification costs reported (\$50,000 estimated)

Why it failed

- Voided by: Form-factor / handling regression

Assessment

The concept fails form factor parity because replacing a single-part caulking gun cartridge with a two-part liquid pumping setup represents a significant operational workflow regression. Per sub-test G, `form_factor_parity=false` forces `like_for_like=false` and `superiority=false`. Additionally, incumbent price parity at \$18.00/kg is explicitly acknowledged as an unsourced proxy.

Also flagged

- Evidence rests on non-peer-reviewed sources
- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-01 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

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